

YASIR AHMED

Digital Twin & IIoT Engineer | Cyber-Physical Systems, Machine Learning, Industrial Automation
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PROFESSIONAL PROFILE

Digital Twin & IIoT Engineer specializing in cyber-physical systems and smart manufacturing. Hands-on expertise in designing real-time data pipelines, commissioning semiconductor manufacturing equipment, and developing predictive maintenance solutions using machine learning. Technical background spanning Lingua Franca, OPC UA integration, enterprise networking, and field service across telecom, VoIP, and industrial automation environments. Available for travel throughout Italy. Open to Digital Twin, IIoT, Field Service, Manufacturing Engineering, or Industry 4.0 roles.

WORK EXPERIENCE

IT Desktop Support Engineer

June 2026 – Onwards

Axiom Technologies • Milan, Italy

- Manage desktop device incidents (hardware and software) through to resolution, delivering IMAC and desktside break-fix support for laptops and desktops.
- Support and maintain desktop configuration, imaging, and patching across the device lifecycle, including SCCM deployment and vendor-coordinated hardware replacements.
- Deliver VIP desktside support and act as the on-site point of contact for IT issues, including Major Incident calls and on-call critical outage support.
- Provide Smart Hands support for network devices and perform endpoint health checks, physical asset audits, and asset manage

Digital Twin & Machine Learning Engineer — Research Internship

Aug 2024 – Oct 2025

University of Verona • Italy

- Designed a Digital Twin architecture for IC production using Lingua Franca, modeling a 7-phase wafer workflow with deterministic event-driven coordination for real-time fault detection.
- Built machine learning pipelines (Random Forest) for anomaly detection on industrial datasets (5,104 wafers, ~32,500 samples), including feature engineering and preprocessing.
- Integrated IIoT sensor streams and equipment interfaces; implemented deterministic timestamping for real-time monitoring ($\leq 500\text{ms}$ latency) and reproducible analytics.
- Developed FastAPI-based data pipelines; conducted sensor calibration, noise reduction, QA stress-testing, and iterative model validation.
- Led root cause analysis of IIoT-detected anomalies, refining models accordingly; produced comprehensive technical documentation and met all project milestones.

VoIP Engineer

Jun 2022 – Sep 2023

Sybrid Private Limited (Remote) • Karachi, Pakistan

- Delivered remote VoIP engineering support to international SME clients, deploying and managing SIP trunk infrastructure, VoIP gateways, and IP telephony endpoints.
- Diagnosed and resolved call-quality degradation, routing bugs, and gateway misconfigurations; maintained SLA compliance across multiple client accounts.

Technical Support Engineer — Network Infrastructure

Jun 2020 – Jul 2021

Enterprise Clients • Karachi, Pakistan

- Provided Tier-2/Tier-3 support for enterprise multi-site WAN/LAN environments and administered Network Management Systems, maintaining 99.9% uptime SLA and driving team-wide topology improvements.

RF Field Intern Engineer

Apr 2019 – Aug 2019

Netkom Technologies (Pvt.) Ltd • Pakistan

- Carried out 2G/3G/4G drive-testing campaigns using TEMS, benchmarking core RF KPIs (call drops, handover success, interference, Rx quality) to guide network optimization decisions.

Engineering Intern — Broadcast/Transmission Engineering

Jan 2019

Radio Pakistan • Karachi, Pakistan

- Gained hands-on exposure to broadcast and transmission engineering systems and operations, including radio signal transmission equipment and broadcast infrastructure workflows.

Engineering Intern — Aircraft Maintenance

Jun 2018 – Jul 2018

Pakistan International Airlines • Karachi, Pakistan

- Completed a 2-week internship with PIA's Engineering & Maintenance division under the HR Training & Automation department, gaining hands-on exposure to aircraft maintenance practices and airworthiness management procedures.

CORE TECHNICAL COMPETENCIES

- Industrial Protocols & Control: OPC UA, PLC (IEC 61131-3 / Structured Text), SCADA, Modbus, Sensor Integration
- Digital Twin & Cyber-Physical Systems: Lingua Franca, Real-Time Simulation, Event-Driven Architecture
- Machine Learning: Python, Scikit-Learn, Random Forest, TensorFlow, Time-Series Anomaly Detection
- Backend & Data Pipelines: FastAPI, REST APIs, SQLite/Time-Series Storage, Git
- Networking & Field Systems: TCP/IP, SIP/VoIP, RF Systems, Network Management Systems
- IT Desktop & Endpoint Support: Windows imaging & SCCM deployment, patch management, deskside/IMAC support, antivirus management, VDI endpoints, asset management

ACADEMIC PROJECTS

Industrial Bottling Line — Cyber-Physical Digital Twin | *Python, OPC UA, FastAPI, Scikit-Learn, Grafana*

- Built an end-to-end digital twin pipeline streaming live OPC UA telemetry from a simulated bottling line into a machine learning fault detector, served via FastAPI and visualized in a real-time Grafana dashboard.

SKaB Digital Twin & Anomaly Detection | *Python, Scikit-Learn, ML*

- Built an industrial anomaly detection framework using SKaB dataset; benchmarked predictive maintenance models on time-series data for fault detection in cyber-physical systems.

Medical Image Segmentation Framework | *TensorFlow, Computer Vision*

- Developed and optimized deep learning segmentation pipelines for medical imagery datasets to improve diagnostic accuracy.

EDUCATION

M.Sc. in Computer Engineering for Robotics and Smart Industry

Graduated: Oct 2025

University of Verona • Verona, Italy.

- Thesis: "The Digital Twin of an IC Production Line Based on Lingua Franca and Machine Learning"—reactive programming, IIoT, and predictive ML.

B.Eng. in Telecommunication Engineering

2016 – 2019

Dawood University of Engineering & Technology • Karachi, Pakistan.

- Thesis: "Scalable IoT Framework for Real-Time Data Acquisition in Smart City Environments." Foundation in signal processing, network protocols, and embedded systems.

HONORS, CERTIFICATIONS & INTERNATIONAL EXPERIENCE

- IoT Microcontrollers: Onboarding a Raspberry Pi Using AWS Greengrass—Amazon Web Services (2026).
- AWS IoT SiteWise—Industrial Asset Modeling & Data Collection—AWS Training and Certification (2026).
- SPACERAISE International Summer School—Merit Scholarship—Gran Sasso Science Institute (May 2026).
- AI for Business—Chamber of Commerce of Verona (2025).
- Mechanical Systems Operator EQF Level 3—GOL Programme / Manpower Srl (2025).
- Health & Safety SSL—D.Lgs. 81/08 (Italy)—(2025).
- International Mobility: Youth Exchange (Poland), Youth Rights Advocate—Council of Europe (Hungary), Professional Development BOLD Programme (Germany).

LANGUAGES & WORK AUTHORIZATION

English C1 – Proficient | Italian A2 – Basic, targeting B1 (improving) | Sindhi C2 – Native

Work Authorization: Resident in Milan; valid Italian Residence Permit. Open to relocate. Available immediately.

Autorizzo il trattamento dei miei dati personali ai sensi del D.Lgs. 196/2003 e dell'art. 13 del GDPR (Regolamento UE 2016/679) ai fini della ricerca e della selezione del personale.